MCO-1D Metachronal Cascade Ornithopter

Flap Drive Fault Analysis

Why the motor stalls above 2 Hz despite a 36 A current limit — and how to fix it in under two minutes

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1. Drive System Description

The flap drive uses a **grooved capstan drum** on the GoolRC 5010 360 KV brushless motor shaft, connected by Dyneema cord to a large **sector wheel** at the wing root. Both ends of the cord are anchored to the sector wheel. The motor drum has machined helical grooves that seat each wrap of cord at a fixed radius, giving a constant 7:1 mechanical reduction throughout the full ±80° flap stroke. The ODrive Micro controller drives the motor over CAN-FD, operating in position control mode with the Raspberry Pi Zero 2W sending smootherstep position commands at 50 Hz.

Drive parameters

Motor KV	360 KV	GoolRC 5010
KT (torque const)	0.02653 Nm/A	From KV: KT = 60/(2π×KV)
Bus voltage	22.2 V (6S LiPo)	Nominal; 25.2 V fully charged
Reduction ratio	7:1	Grooved drum on motor → sector wheel
Stroke amplitude	±80°	Wing root
Max flap frequency	3.0 Hz	Design target
Current limit set	**36 A**	Well above the 14.9 A peak demand — not the problem

2. The Symptom

The motor operates correctly below approximately 2 Hz. Above that threshold it stalls, overheats, and fails to maintain the commanded position profile. The problem persists even after raising **current_lim to 36 A** — well above the calculated peak demand of 14.9 A at 3 Hz.

> ■ **Motor stalls and overheats above ~2 Hz despite 36 A current limit. Current limit is NOT the bottleneck.**

3. Why Current and Back-EMF Are Not the Cause

Two alternative explanations can be ruled out immediately by calculation.

Back-EMF at peak motor speed

Flap freq	Motor peak (rad/s)	Motor peak (RPM)	Back-EMF (V)	% of 22.2 V supply	Current needed (A)
1.0 Hz	37	350	0.97	4.4%	1.7
1.5 Hz	55	525	1.46	6.6%	3.7
2.0 Hz	73	700	1.94	8.8%	6.6

2.5 Hz	92	875	2.43	10.9%	10.4
3.0 Hz	110	1050	2.92	13.1%	14.9

✓ Back-EMF at 3 Hz is only 2.9 V — 13% of the 22.2 V supply. The motor has 87% of its voltage headroom available at all operating frequencies. Back-EMF is not the cause.

Current demand at 3 Hz is 14.9 A — less than half the 36 A limit. Current is not the cause either. The motor has abundant electrical headroom. The fault must be elsewhere in the ODrive configuration.

4. The Most Likely Cause: vel_limit

The ODrive **vel_limit** parameter sets the maximum motor shaft angular velocity the controller will allow. If the motor exceeds this value during a stroke — even briefly at the mid-stroke velocity peak — the ODrive triggers **ERROR_VELOCITY_LIMIT_VIOLATION** and cuts all motor output instantly. The sector wheel then coasts freely on its own inertia, the cord goes momentarily slack, and when the ODrive recovers it fights hard to regain position — producing a large current spike and significant heat. This sequence repeats six times per second at 3 Hz.

The smootherstep velocity profile peaks at **1.875 × amplitude / stroke_duration** at mid-stroke. Through the 7:1 reduction, the motor shaft peaks at:

Flap freq	Motor shaft peak (rad/s)	Motor shaft peak (RPM)	vel_limit needed (rad/s)	Status if vel_limit = 80
1.0 Hz	37	350	≥ 42	**OK ✓**
1.5 Hz	55	525	≥ 63	**OK ✓**
2.0 Hz	73	700	≥ 84	**OK ✓**
2.5 Hz	**92**	875	≥ 105	**FAULT ✗**
3.0 Hz	**110**	1050	≥ 126	**FAULT ✗**

At which frequency does the motor exceed a given vel_limit?

vel_limit setting (rad/s)	Motor exceeds it above (Hz)	Explains 2 Hz failure?
20	0.55 Hz	—
40	1.09 Hz	—
60	**1.64 Hz**	**YES — prime suspect**
80	**2.18 Hz**	**YES — prime suspect**
100	2.73 Hz	—
120	3.27 Hz	—
150	4.09 Hz	—

■ A vel_limit near 80 rad/s would cause ERROR_VELOCITY_LIMIT_VIOLATION at exactly 2.2 Hz — matching the observed failure threshold precisely. The ODrive default in many firmware versions is far below 150 rad/s.

5. Diagnostic — Run This Immediately After a Failed Attempt

Without clearing the ODrive state after a failure, read the error registers. The error code identifies the fault path exactly.

```
# In odrivetool — run immediately after motor stalls at high frequency
print(odrv0.axis0.motor.error) # 0 = no motor fault
print(odrv0.axis0.controller.error) # 0 = no controller fault
print(odrv0.axis0.error) # 0 = no axis fault

# Then check the current vel_limit setting:
print(odrv0.axis0.controller.config.vel_limit)
```

Error code returned	Meaning
ERROR_VELOCITY_LIMIT_VIOLATION	vel_limit is the problem — apply the fix in Section 6
ERROR_CURRENT_LIMIT_VIOLATION	Current is somehow still the issue — check requested_current_range
All zeros, motor stopped	Fault is mechanical — check for binding in the drive linkage

6. The Fix — Two Minutes in odrivetool

If ERROR_VELOCITY_LIMIT_VIOLATION is confirmed, apply the following. The required motor shaft velocity at 3 Hz is 110 rad/s. Setting vel_limit to 150 rad/s gives a 1.4× safety margin.

```
# ■■ Apply these settings in odrivetool ■■■■■■■■■■■■■■■■■■■■■■■■

# Primary fix: velocity limit
odrv0.axis0.controller.config.vel_limit = 150 # rad/s
odrv0.axis0.controller.config.vel_limit_tolerance = 16 # rad/s

# Verify while here:
odrv0.axis0.motor.config.pole_pairs = 7 # CRITICAL
odrv0.axis0.motor.config.current_control_bandwidth = 2000 # rad/s

# Save and reboot
odrv0.save_configuration()
odrv0.reboot()

# After reboot — verify no errors
odrv0.axis0.requested_state = AXIS_STATE_CLOSED_LOOP_CONTROL
print(odrv0.axis0.motor.error) # expect 0
print(odrv0.axis0.controller.error) # expect 0
```

✓ These two lines alone — vel_limit = 150 and vel_limit_tolerance = 16 — will very likely resolve the 2 Hz ceiling immediately, without any mechanical changes to the drive.

7. Additional Settings to Verify

While in odrivetool, verify the following parameters. Any one of these being wrong can cause similar symptoms independently of vel_limit.

Parameter	Required value	If wrong — consequence
`motor.config.pole_pairs`	**7**	Wrong commutation at speed → weak torque → stall. Count rotor magnets ÷ 2.
`motor.config.current_control_bandwidth`	**2000 rad/s**	Inner loop too slow to follow 3 Hz demand → apparent current shortfall despite high limit.
`motor.config.phase_resistance (after calib)`	0.05–0.09 Ω	Outside range → calibration failed → wrong PI gains → oscillation and heat.
`motor.config.phase_inductance (after calib)`	10–25 µH	Near zero → calibration_current was too low. Increase to 8 A and re-calibrate.
`config.dc_max_negative_current`	–25.0 A	If 0 or not set: regenerative braking current at stroke reversal has no path → bus voltage spike → ODrive faults.

8. Why the Motor Overheats When vel_limit is Too Low

When the ODrive triggers ERROR_VELOCITY_LIMIT_VIOLATION it cuts all motor output **instantly**. The large sector wheel, carrying the wing's kinetic energy, continues rotating under inertia. The cord on the return side goes momentarily slack. The ODrive then attempts to recover the position error by commanding **maximum current** — which is now 36 A. This brief burst of maximum current generates I^2R heat in all three motor phases. At 3 Hz this occurs six times per second. Even brief 36 A pulses, repeated continuously, are sufficient to heat the stator rapidly.

This is why the motor runs hot despite the mechanical drive being correctly designed and the current limit being raised. The heat is not from sustained high current during normal operation — it is from the recovery current spikes that follow each velocity limit violation.